Tutorial – 6

1. Write a program to define an interface IEmployee with methods to accept and display employee details. Implement the interface using an Employee class. (Interface)

```
2.
3.  using System;
4.
5.  namespace _24SOECE11019_RAJVI_LUNAGARIYA_TUTORIAL_6
6.  {
7.      interface IEmployee
8.      {
9.          void AcceptDetails();
10.         void DisplayDetails();
11.     }
12.
13.     class Employee : IEmployee
14.     {
15.         int id;
16.         string name;
17.         double salary;
18.
19.         public void AcceptDetails()
20.         {
21.             Console.Write("Enter Employee ID: ");
22.             id = Convert.ToInt32(Console.ReadLine());
23.
24.             Console.Write("Enter Employee Name: ");
25.             name = Console.ReadLine();
26.
27.             Console.Write("Enter Employee Salary: ");
28.             salary = Convert.ToDouble(Console.ReadLine());
29.         }
30.
31.         public void DisplayDetails()
32.         {
33.             Console.WriteLine("\n--- Employee Details ---");
34.             Console.WriteLine("Employee ID: " + id);
35.             Console.WriteLine("Employee Name: " + name);
36.             Console.WriteLine("Employee Salary: " + salary);
37.         }
38.     }
39.
40.     class Program1
41.     {
42.         static void Main(string[] args)
43.         {
44.             Employee emp = new Employee();
45.
46.             emp.AcceptDetails();
```

```
            emp.DisplayDetails();

            Console.ReadLine();
        }
    }
}
```

```
D:\.net Project\24SOECE11019_RAJVI_LUNAGARIYA_TUTORIAL-6>IEmployee
Enter Employee ID: 4
Enter Employee Name: rajvi
Enter Employee Salary: 20000

--- Employee Details ---
Employee ID: 4
Employee Name: rajvi
Employee Salary: 20000
```

2. Write a program to define two interfaces, IPersonalDetails and IProfessionalDetails, and implement both interfaces in an Employee class to display complete employee information.(Multiple interfaces)

```
using System;

namespace _24SOECE11019_RAJVI_LUNAGARIYA_TUTORIAL_6
{
    interface IPersonalDetails
    {
        void AcceptPersonalDetails();
        void DisplayPersonalDetails();
    }

    interface IProfessionalDetails
    {
        void AcceptProfessionalDetails();
        void DisplayProfessionalDetails();
    }

    class Employee2 : IPersonalDetails, IProfessionalDetails
    {
        int id;
        string name = "";
        string address = "";
        string designation = "";
        double salary;
```

```
    public void AcceptPersonalDetails()
    {
        Console.Write("Enter Employee ID: ");
        id = Convert.ToInt32(Console.ReadLine());

        Console.Write("Enter Employee Name: ");
        name = Console.ReadLine();

        Console.Write("Enter Employee Address: ");
        address = Console.ReadLine();
    }

    public void AcceptProfessionalDetails()
    {
        Console.Write("Enter Employee Designation: ");
        designation = Console.ReadLine();

        Console.Write("Enter Employee Salary: ");
        salary = Convert.ToDouble(Console.ReadLine());
    }

    public void DisplayPersonalDetails()
    {
        Console.WriteLine("\n--- Personal Details ---");
        Console.WriteLine("Employee ID: " + id);
        Console.WriteLine("Employee Name: " + name);
        Console.WriteLine("Employee Address: " + address);
    }

    public void DisplayProfessionalDetails()
    {
        Console.WriteLine("\n--- Professional Details ---");
        Console.WriteLine("Designation: " + designation);
        Console.WriteLine("Salary: " + salary);
    }
}

class Program2
{
    static void Main(string[] args)
    {
        Employee2 emp = new Employee2();

        emp.AcceptPersonalDetails();
```

```
            emp.AcceptProfessionalDetails();

            emp.DisplayPersonalDetails();
            emp.DisplayProfessionalDetails();

            Console.ReadLine();
        }
    }
}
```

```
D:\.net Project\24SOECE11019_RAJVI_LUNAGARIYA_TUTORIAL-6>IPersonalDetails
Enter Employee ID: 4
Enter Employee Name: rajvi
Enter Employee Address: morbi road
Enter Employee Designation: ...
Enter Employee Salary: 20000

--- Personal Details ---
Employee ID: 4
Employee Name: rajvi
Employee Address: morbi road

--- Professional Details ---
Designation: ...
Salary: 20000
```

3. Write a program to create a delegate Calculate which accepts two numbers and performs addition, subtraction, multiplication and division using different methods.(Delegate)

```
using System;

namespace _24SOECE11019_RAJVI_LUNAGARIYA_TUTORIAL_6
{
    delegate double Calculate(double a, double b);

    class Program3
    {
        static double Addition(double a, double b)
        {
            return a + b;
```

```
}

static double Subtraction(double a, double b)
{
   return a - b;
}

static double Multiplication(double a, double b)
{
   return a * b;
}

static double Division(double a, double b)
{
   return a / b;
}

static void Main(string[] args)
{
   Console.Write("Enter first number: ");
   double num1 = Convert.ToDouble(Console.ReadLine());

   Console.Write("Enter second number: ");
   double num2 = Convert.ToDouble(Console.ReadLine());

   Calculate calc;

   calc = Addition;
   Console.WriteLine("Addition: " + calc(num1, num2));

   calc = Subtraction;
   Console.WriteLine("Subtraction: " + calc(num1, num2));

   calc = Multiplication;
   Console.WriteLine("Multiplication: " + calc(num1, num2));

   if (num2 != 0)
```

```
            {
                calc = Division;
                Console.WriteLine("Division: " + calc(num1, num2));
            }
            else
            {
                Console.WriteLine("Division by zero is not possible.");
            }

            Console.ReadLine();
        }
    }
}
```

```
D:\.net Project\24SOECE11019_RAJVI_LUNAGARIYA_TUTORIAL-6>program3
Enter first number: 1
Enter second number: 3
Addition: 4
Subtraction: -2
Multiplication: 3
Division: 0.333333333333333
```

4. Write a program to demonstrate a multicast delegate by creating a delegate that calls multiple methods to display student information such as name, course and semester. (Multicast Delegate)

```
using System;

namespace _24SOECE11019_RAJVI_LUNAGARIYA_TUTORIAL_6
{
    delegate void StudentInfo();

    class Program4
    {
        static void DisplayName()
        {
```

```
            Console.WriteLine("Student Name: Rajvi Lunagariya");
        }

        static void DisplayCourse()
        {
            Console.WriteLine("Course: B.Tech");
        }

        static void DisplaySemester()
        {
            Console.WriteLine("Semester: 4th Semester");
        }

        static void Main(string[] args)
        {
            StudentInfo student;

            student = DisplayName;
            student += DisplayCourse;
            student += DisplaySemester;

            Console.WriteLine("--- Student Information ---");
            student();

            Console.ReadLine();
        }
    }
}
```

```
D:\.net Project\24SOECE11019_RAJVI_LUNAGARIYA_TUTORIAL-6>Program4
--- Student Information ---
Student Name: Rajvi Lunagariya
Course: B.Tech
Semester: 4th Semester
```

5. Write a program to demonstrate event handling by creating a BankAccount class. Raise an event whenever money is deposited into the account and display an appropriate message.(Events)

```
using System;

namespace _24SOECE11019_RAJVI_LUNAGARIYA_TUTORIAL_6
{
   class BankAccount
   {
      public event EventHandler MoneyDeposited;

      public double Balance { get; private set; }

      public void Deposit(double amount)
      {
         if (amount <= 0)
         {
            Console.WriteLine("Please enter a valid deposit amount.");
            return;
         }

         Balance += amount;

         MoneyDeposited?.Invoke(this, EventArgs.Empty);
      }

      public void DisplayMessage(object sender, EventArgs e)
      {
         Console.WriteLine("Money deposited successfully!");
         Console.WriteLine("Current Balance: " + Balance);
      }
   }

   class Program5
   {
```

```
        static void Main(string[] args)
        {
            BankAccount account = new BankAccount();

            account.MoneyDeposited += account.DisplayMessage;

            Console.Write("Enter deposit amount: ");
            double amount = Convert.ToDouble(Console.ReadLine());

            account.Deposit(amount);

            Console.ReadLine();
        }
    }
}
```

```
D:\.net Project\24SOECE11019_RAJVI_LUNAGARIYA_TUTORIAL-6>BankAccount
Enter deposit amount: 2000
Money deposited successfully!
Current Balance: 2000
```

6. Write a program which accepts two numbers from the user and performs division. Handle the DivideByZeroException using a try-catch block.(Exception)

```
using System;

namespace _24SOECE11019_RAJVI_LUNAGARIYA_TUTORIAL_6
{
    class Program6
    {
        static void Main(string[] args)
        {
            try
            {
                Console.Write("Enter first number: ");
                int num1 = Convert.ToInt32(Console.ReadLine());
```

```
            Console.Write("Enter second number: ");
            int num2 = Convert.ToInt32(Console.ReadLine());

            int result = num1 / num2;

            Console.WriteLine("Division Result: " + result);
        }
        catch (DivideByZeroException)
        {
            Console.WriteLine("Error: Cannot divide by zero.");
        }

        Console.ReadLine();
    }
  }
}
```

```
D:\.net Project\24SOECE11019_RAJVI_LUNAGARIYA_TUTORIAL-6>Program6
Enter first number: 4
Enter second number: 2
Division Result: 2
```

7. Write a program which accepts an integer from the user and handles the exception when the user enters an invalid value instead of a number. Use appropriate try-catch-finally blocks.(Try-Catch-Finally)

```
using System;

namespace _24SOECE11019_RAJVI_LUNAGARIYA_TUTORIAL_6
{
  class Program7
  {
    static void Main(string[] args)
    {
```

```
            try
            {
                Console.Write("Enter an integer: ");
                int num = Convert.ToInt32(Console.ReadLine());

                Console.WriteLine("You entered: " + num);
            }
            catch (FormatException)
            {
                Console.WriteLine("Error: Please enter a valid integer.");
            }
            finally
            {
                Console.WriteLine("Finally block executed.");
            }

            Console.ReadLine();
        }
    }
}
```

```
D:\.net Project\24SOECE11019_RAJVI_LUNAGARIYA_TUTORIAL-6>Program7
Enter an integer: 4
You entered: 4
Finally block executed.
```

8. Write a program to create a custom exception InsufficientBalanceException for a bank account. If the withdrawal amount is greater than the available balance, generate the custom exception.(Custom Exception)

```
using System;

namespace _24SOECE11019_RAJVI_LUNAGARIYA_TUTORIAL_6
{
    // Custom Exception
    class InsufficientBalanceException : Exception
```

```
{
    public InsufficientBalanceException(string message) : base(message)
    {
    }
}

// Bank Account Class
class BankAccount1
{
    double balance = 50000;

    public void Withdraw(double amount)
    {
        if (amount > balance)
        {
            throw new InsufficientBalanceException(
                "Error: Insufficient balance for withdrawal.");
        }

        balance -= amount;

        Console.WriteLine("Withdrawal successful.");
        Console.WriteLine("Remaining Balance: " + balance);
    }
}

class Program8
{
    static void Main(string[] args)
    {
        BankAccount1 account = new BankAccount1();

        try
        {
            Console.Write("Enter amount to withdraw: ");
            double amount = Convert.ToDouble(Console.ReadLine());
```

```
            account.Withdraw(amount);
        }
        catch (InsufficientBalanceException ex)
        {
            Console.WriteLine(ex.Message);
        }

        Console.ReadLine();
    }
  }
}
```

```
D:\.net Project\24SOECE11019_RAJVI_LUNAGARIYA_TUTORIAL-6>Program8
Enter amount to withdraw: 4000
Withdrawal successful.
Remaining Balance: 46000
```

9. Write a program which accepts two numbers from the user and uses a lambda expression to calculate their addition, multiplication and maximum value.
(Lambda)

```
using System;

namespace _24SOECE11019_RAJVI_LUNAGARIYA_TUTORIAL_6
{
  class Program9
  {
    static void Main(string[] args)
    {
      Console.Write("Enter first number: ");
      int a = Convert.ToInt32(Console.ReadLine());

      Console.Write("Enter second number: ");
      int b = Convert.ToInt32(Console.ReadLine());
```

```
            // Lambda expressions
            Func<int, int, int> addition = (x, y) => x + y;
            Func<int, int, int> multiplication = (x, y) => x * y;
            Func<int, int, int> maximum = (x, y) => x > y ? x : y;

            Console.WriteLine("Addition: " + addition(a, b));
            Console.WriteLine("Multiplication: " + multiplication(a, b));
            Console.WriteLine("Maximum: " + maximum(a, b));

            Console.ReadLine();
        }
    }
}
```

```
D:\.net Project\24SOECE11019_RAJVI_LUNAGARIYA_TUTORIAL-6>Program9
Enter first number: 5
Enter second number: 6
Addition: 11
Multiplication: 30
Maximum: 6
```

10. Write a program to create a list of student marks and use a lambda expression to display students who have obtained marks greater than 50.(Lambda + Collection)

```
using System;
using System.Collections.Generic;

namespace _24SOECE11019_RAJVI_LUNAGARIYA_TUTORIAL_6
{
    class Program10
    {
        static void Main(string[] args)
        {
            List<int> marks = new List<int> { 45, 75, 60, 35, 90, 50, 85 };
```

```
            Console.WriteLine("Student Marks Greater Than 50:");

            marks.FindAll(m => m > 50)
                .ForEach(m => Console.WriteLine(m));

            Console.ReadLine();
        }
    }
}
```

```
D:\.net Project\24SOECE11019_RAJVI_LUNAGARIYA_TUTORIAL-6>Program10
Student Marks Greater Than 50:
75
60
90
85
```

11. Write a program to create a collection of integers and use LINQ to display all even numbers from the collection in ascending order.(LINQ)

```
using System;
using System.Collections.Generic;
using System.Linq;

namespace _24SOECE11019_RAJVI_LUNAGARIYA_TUTORIAL_6
{
    class Program11
    {
        static void Main(string[] args)
        {
            List<int> numbers = new List<int>
            {
                15, 8, 3, 12, 6, 21, 10, 4, 18
            };
```

```
            var evenNumbers = from n in numbers
                              where n % 2 == 0
                              orderby n ascending
                              select n;

            Console.WriteLine("Even numbers in ascending order:");

            foreach (int n in evenNumbers)
            {
                Console.WriteLine(n);
            }

            Console.ReadLine();
        }
    }
}
```

```
D:\.net Project\24SOECE11019_RAJVI_LUNAGARIYA_TUTORIAL-6>Program11
Even numbers in ascending order:
4
6
8
10
12
18
```

12. Write a program to create a list of Student objects containing Student ID, Name, Course and Marks. Use LINQ to display students whose marks are greater than 70 and sort them by marks.(LINQ + Objects)

```
using System;
using System.Collections.Generic;
using System.Linq;

namespace _24SOECE11019_RAJVI_LUNAGARIYA_TUTORIAL_6
{
    class Student
    {
```

```
        public int StudentID { get; set; }
        public string Name { get; set; }
        public string Course { get; set; }
        public double Marks { get; set; }
    }

    class Program12
    {
        static void Main(string[] args)
        {
            List<Student> students = new List<Student>
            {
                new Student { StudentID = 1, Name = "Rajvi", Course = "B.Tech",
Marks = 85 },
                new Student { StudentID = 2, Name = "Amit", Course = "BCA", Marks
= 65 },
                new Student { StudentID = 3, Name = "Priya", Course = "B.Tech",
Marks = 92 },
                new Student { StudentID = 4, Name = "Rahul", Course = "BCA", Marks
= 75 },
                new Student { StudentID = 5, Name = "Neha", Course = "B.Tech",
Marks = 60 }
            };

            var result = from s in students
                         where s.Marks > 70
                         orderby s.Marks ascending
                         select s;

            Console.WriteLine("Students with Marks Greater Than 70:");
            Console.WriteLine("-----------------------------------");

            foreach (Student s in result)
            {
                Console.WriteLine("Student ID: " + s.StudentID);
                Console.WriteLine("Name: " + s.Name);
                Console.WriteLine("Course: " + s.Course);
```

```
            Console.WriteLine("Marks: " + s.Marks);
            Console.WriteLine();
        }

        Console.ReadLine();
    }
  }
}
```

```
D:\.net Project\24SOECE11019_RAJVI_LUNAGARIYA_TUTORIAL-6>Program12
Students with Marks Greater Than 70:
-----------------------------------
Student ID: 4
Name: Rahul
Course: BCA
Marks: 75

Student ID: 1
Name: Rajvi
Course: B.Tech
Marks: 85

Student ID: 3
Name: Priya
Course: B.Tech
Marks: 92
```

13. Write a program to create and start multiple threads to perform different tasks such as displaying numbers from 1 to 10, displaying alphabets from A to J, and displaying a message.(Multithreading)

```
using System;
using System.Threading;

namespace _24SOECE11019_RAJVI_LUNAGARIYA_TUTORIAL_6
{
  class Program13
  {
    static void DisplayNumbers()
    {
      for (int i = 1; i <= 10; i++)
```

```
        {
            Console.WriteLine("Number: " + i);
            Thread.Sleep(100);
        }
    }

    static void DisplayAlphabets()
    {
        for (char ch = 'A'; ch <= 'J'; ch++)
        {
            Console.WriteLine("Alphabet: " + ch);
            Thread.Sleep(100);
        }
    }

    static void DisplayMessage()
    {
        for (int i = 1; i <= 5; i++)
        {
            Console.WriteLine("Hello! This is a separate thread.");
            Thread.Sleep(100);
        }
    }

    static void Main(string[] args)
    {
        Thread t1 = new Thread(DisplayNumbers);
        Thread t2 = new Thread(DisplayAlphabets);
        Thread t3 = new Thread(DisplayMessage);

        t1.Start();
        t2.Start();
        t3.Start();

        t1.Join();
        t2.Join();
        t3.Join();
```

```
            Console.WriteLine("All threads completed.");

            Console.ReadLine();
        }
    }
}
```

```
D:\.net Project\24SOECE11019_RAJVI_LUNAGARIYA_TUTORIAL-6>Program13
Alphabet: A
Hello! This is a separate thread.
Number: 1
Hello! This is a separate thread.
Number: 2
Alphabet: B
Alphabet: C
Hello! This is a separate thread.
Number: 3
Number: 4
Hello! This is a separate thread.
Alphabet: D
Hello! This is a separate thread.
Alphabet: E
Number: 5
Number: 6
Alphabet: F
Alphabet: G
Number: 7
Alphabet: H
Number: 8
Alphabet: I
Number: 9
Alphabet: J
Number: 10
All threads completed.
```

14. Write a program to demonstrate the thread lifecycle by creating a thread, starting it, checking its state, waiting for its completion using Join(), and displaying the thread status at different stages.(Thread Lifecycle)

```
using System;
using System.Threading;

namespace _24SOECE11019_RAJVI_LUNAGARIYA_TUTORIAL_6
{
```

```
class Program14
{
    static void DisplayMessage()
    {
        Console.WriteLine("Thread is running...");

        Thread.Sleep(2000);

        Console.WriteLine("Thread task completed.");
    }

    static void Main(string[] args)
    {
        Thread t = new Thread(DisplayMessage);

        Console.WriteLine("Thread Created.");
        Console.WriteLine("Thread State: " + t.ThreadState);

        t.Start();

        Console.WriteLine("Thread Started.");
        Console.WriteLine("Thread State: " + t.ThreadState);

        Console.WriteLine("Waiting for thread to complete...");

        t.Join();

        Console.WriteLine("Thread Execution Finished.");
        Console.WriteLine("Thread State: " + t.ThreadState);

        Console.ReadLine();
    }
}
}
```

```
D:\.net Project\24SOECE11019_RAJVI_LUNAGARIYA_TUTORIAL-6>Program14
Thread Created.
Thread State: Unstarted
Thread Started.
Thread is running...
Thread State: Running
Waiting for thread to complete...
Thread task completed.
Thread Execution Finished.
Thread State: Stopped
```